

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

Perihal Produk: **4-Bromoaniline**  
 Product Description: **4-Bromoaniline**  
 Cat No. : 146280000; 146280050; 146280250; 146281000  
 Synonyms p-Bromoaniline; p-bromophenylamine.  
 CAS No 106-40-1  
 Molecular Formula C6 H6 Br N

**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use Laboratory chemicals.  
 Uses advised against No Information available

**Company**

Thermo Fisher Scientific Fisher Scientific (M) Sdn Bhd  
 Hap Seng Business Park, Lot 01-03, 01-04 Aras 1 Unity Square,  
 No 12, Persiaran Perusahaan, Seksyen 23, 40300 Shah Alam,  
 Selangor Darul Ehsan, Malaysia.  
 Main line: +60 3-5525 7888

**E-mail address**

Enquiry.my@thermofisher.com

**Emergency Telephone Number**

Tel: +03-5525 7888  
 CHEMTREC Malaysia **1-800-815-308** (Malay)  
 CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture**

|                                                      |                   |
|------------------------------------------------------|-------------------|
| Acute oral toxicity                                  | Category 4 (H302) |
| Acute dermal toxicity                                | Category 3 (H311) |
| Acute Inhalation Toxicity - Dusts and Mists          | Category 4 (H332) |
| Skin Corrosion/Irritation                            | Category 2 (H315) |
| Serious Eye Damage/Eye Irritation                    | Category 2 (H319) |
| Specific target organ toxicity - (repeated exposure) | Category 2 (H373) |

**Label Elements**


Signal Word

Danger

# SAFETY DATA SHEET

4-Bromoaniline

Revision Date 21-Mar-2025

## Hazard Statements

H311 - Toxic in contact with skin  
H319 - Causes serious eye irritation  
H315 - Causes skin irritation  
H373 - May cause damage to organs through prolonged or repeated exposure  
H302 + H332 - Harmful if swallowed or if inhaled

## Precautionary Statements

### Prevention

P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

### Response

P302 + P352 - IF ON SKIN: Wash with plenty of soap and water  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P312 - Call a POISON CENTER or doctor if you feel unwell  
P330 - Rinse mouth  
P361 + P364 - Take off immediately all contaminated clothing and wash it before reuse

### Storage

P405 - Store locked up

### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Other Hazards

Toxic to terrestrial vertebrates  
This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component             | CAS No   | Weight % |
|-----------------------|----------|----------|
| Benzenamine, 4-bromo- | 106-40-1 | >95      |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### General Advice

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.

#### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

#### Ingestion

Do NOT induce vomiting. Call a physician or poison control center immediately.

#### Inhalation

Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a

# SAFETY DATA SHEET

4-Bromoaniline

Revision Date 21-Mar-2025

pocket mask equipped with a one-way valve or other proper respiratory medical device.  
Immediate medical attention is required.

**Self-Protection of the First Aider** Use personal protective equipment as required.

**Most important symptoms and effects, both acute and delayed**

None reasonably foreseeable.

**Indication of any immediate medical attention and special treatment needed**

**Notes to Physician** Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

**Extinguishing media**

**Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Special hazards arising from the substance or mixture**

Thermal decomposition can lead to release of irritating gases and vapors.

**Hazardous Combustion Products**

Nitrogen oxides (NO<sub>x</sub>), Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Hydrogen halides.

**Advice for fire-fighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

**Personal Precautions, Protective Equipment and Emergency Procedures**

Use personal protective equipment as required. Ensure adequate ventilation. Avoid dust formation. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas.

**Environmental precautions**

Should not be released into the environment. See Section 12 for additional Ecological Information. Do not flush into surface water or sanitary sewer system.

**Methods and Material for Containment and Cleaning Up**

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

**Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

**Precautions for Safe Handling**

# SAFETY DATA SHEET

4-Bromoaniline

Revision Date 21-Mar-2025

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Avoid dust formation. Use only under a chemical fume hood. Do not breathe (dust, vapor, mist, gas). Do not ingest. If swallowed then seek immediate medical assistance.

## **Conditions for Safe Storage, Including any Incompatibilities**

Keep containers tightly closed in a dry, cool and well-ventilated place. Store under an inert atmosphere. Protect from direct sunlight.

## **Specific End Uses**

Use in laboratories.

## **SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

### **Control Parameters**

### **Exposure Controls**

#### **Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### **Personal protective equipment**

#### **Eye Protection**

Goggles

#### **Hand Protection**

Protective gloves

#### **Skin and body protection**

Long sleeved clothing

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

#### **Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators

#### **Recommended Filter type:**

Particulates filter conforming to EN 143

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

When RPE is used a face piece Fit Test should be conducted

### **Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice

### **Environmental exposure controls**

Prevent product from entering drains

## **SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES**

### **Information on basic physical and chemical properties**

# SAFETY DATA SHEET

4-Bromoaniline

Revision Date 21-Mar-2025

|                                         |                             |                                   |
|-----------------------------------------|-----------------------------|-----------------------------------|
| Appearance                              | Grey                        |                                   |
| Physical State                          | Solid                       |                                   |
| Odor                                    | Strong                      |                                   |
| Odor Threshold                          | No data available           |                                   |
| pH                                      | 3.7-4.0                     |                                   |
| Melting Point/Range                     | 60 - 64 °C / 140 - 147.2 °F |                                   |
| Softening Point                         | No data available           |                                   |
| Boiling Point/Range                     | 230 - 250 °C / 446 - 482 °F |                                   |
| Flash Point                             | 110 °C / 230 °F             | Method - No information available |
| Evaporation Rate                        | Not applicable              | Solid                             |
| Flammability (solid,gas)                | No information available    |                                   |
| Explosion Limits                        | No data available           |                                   |
| Vapor Pressure                          | 0.23 mbar @ 25 °C           |                                   |
| Vapor Density                           | Not applicable              | Solid                             |
| Specific Gravity / Density              | 1.497                       |                                   |
| Bulk Density                            | No data available           |                                   |
| Water Solubility                        | < 1 g / L                   |                                   |
| Solubility in other solvents            | No information available    |                                   |
| Partition Coefficient (n-octanol/water) |                             |                                   |
| Component                               | log Pow                     |                                   |
| Benzenamine, 4-bromo-                   | 2.32                        |                                   |
| Autoignition Temperature                | No data available           |                                   |
| Decomposition Temperature               | > 230°C                     |                                   |
| Viscosity                               | Not applicable              | Solid                             |
| Explosive Properties                    | No information available    |                                   |
| Oxidizing Properties                    | No information available    |                                   |
| Molecular Formula                       | C6 H6 Br N                  |                                   |
| Molecular Weight                        | 172.02                      |                                   |

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

None known, based on information available.

### Chemical Stability

Sensitivity to light. Air sensitive.

### Possibility of Hazardous Reactions

|                          |                                          |
|--------------------------|------------------------------------------|
| Hazardous Polymerization | Hazardous polymerization does not occur. |
| Hazardous Reactions      | None under normal processing.            |

### Conditions to Avoid

# SAFETY DATA SHEET

4-Bromoaniline

Revision Date 21-Mar-2025

Incompatible products. Excess heat. Exposure to light. Exposure to air.

## Incompatible Materials

Acids. Acid anhydrides. Acid chlorides. Chloroformates.

## Hazardous Decomposition Products

Nitrogen oxides (NO<sub>x</sub>). Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Hydrogen halides.

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### Product Information

##### (a) acute toxicity;

|            |            |
|------------|------------|
| Oral       | Category 4 |
| Dermal     | Category 3 |
| Inhalation | Category 4 |

| Component             | LD50 Oral                | LD50 Dermal       | LC50 Inhalation |
|-----------------------|--------------------------|-------------------|-----------------|
| Benzenamine, 4-bromo- | LD50 = 456 mg/kg ( Rat ) | 536 mg/kg ( Rat ) | -               |

(b) skin corrosion/irritation; Category 2

(c) serious eye damage/irritation; Category 2

##### (d) respiratory or skin sensitization;

|             |                   |
|-------------|-------------------|
| Respiratory | No data available |
| Skin        | No data available |

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available

There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; Category 2

Target Organs Blood, Hematopoietic System.

(j) aspiration hazard; Not applicable  
Solid

# SAFETY DATA SHEET

4-Bromoaniline

Revision Date 21-Mar-2025

**Other Adverse Effects** The toxicological properties have not been fully investigated.

**Symptoms / effects, both acute and delayed** No information available.

**Endocrine Disrupting Properties** Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

**Ecotoxicity effects** Contains a substance which is: Harmful to aquatic organisms. The product contains following substances which are hazardous for the environment.

| Component             | Freshwater Fish                                           | Water Flea | Freshwater Algae | Microtox |
|-----------------------|-----------------------------------------------------------|------------|------------------|----------|
| Benzenamine, 4-bromo- | LC50: = 47.5 mg/L, 96h flow-through (Pimephales promelas) |            |                  |          |

### Persistence and degradability

**Persistence**

Persistence is unlikely.

**Degradation in sewage treatment plant**

Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

### Bioaccumulative potential

Bioaccumulation is unlikely

| Component             | log Pow | Bioconcentration factor (BCF) |
|-----------------------|---------|-------------------------------|
| Benzenamine, 4-bromo- | 2.32    | No data available             |

### Mobility in soil

The product is insoluble and sinks in water. The product is water soluble, and may spread in water systems. Is not likely mobile in the environment due its low water solubility. Will likely be mobile in the environment due to its water solubility. Highly mobile in soils.

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors

**Other adverse effects** No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

### Waste treatment methods

**Waste from Residues/Unused Products**

Waste is classified as hazardous Dispose of in accordance with the European Directives on waste and hazardous waste Dispose of in accordance with local regulations

### Contaminated Packaging

Dispose of this container to hazardous or special waste collection point.

### Other Information

Do not flush to sewer Waste codes should be assigned by the user based on the application for which the product was used Do not empty into drains

## SECTION 14: TRANSPORT INFORMATION

**IMDG/IMO**  
UN-No

UN2811

ACR14628

# SAFETY DATA SHEET

4-Bromoaniline

Revision Date 21-Mar-2025

**Hazard Class** 6.1  
**Packing Group** III  
**Proper Shipping Name** Toxic solid, organic, n.o.s. (4-BROMOANILINE)

## Road and Rail Transport

**UN-No** UN2811  
**Hazard Class** 6.1  
**Packing Group** III  
**Proper Shipping Name** Toxic solid, organic, n.o.s. (4-BROMOANILINE)

## IATA

**UN-No** UN2811  
**Hazard Class** 6.1  
**Packing Group** III  
**Proper Shipping Name** TOXIC SOLID, ORGANIC, N.O.S.\* (4-BROMOANILINE)

**Special Precautions for User** No special precautions required

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

**International Inventories** X = listed

| Component             | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL     |
|-----------------------|-----------|------|-----|-------|------|------|-------|------|----------|
| Benzenamine, 4-bromo- | 203-393-9 | X    | X   | X     | X    | X    | X     | X    | KE-03623 |

### National Regulations

**Persistent Organic Pollutant** This product does not contain any known or suspected substance  
**Ozone Depletion Potential** This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%



# SAFETY DATA SHEET

4-Bromoaniline

Revision Date 21-Mar-2025

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Revision Date

21-Mar-2025

Revision Summary

Not applicable.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**